Estimation Theory

Best Linear Unbiased Estimators (BLUE)

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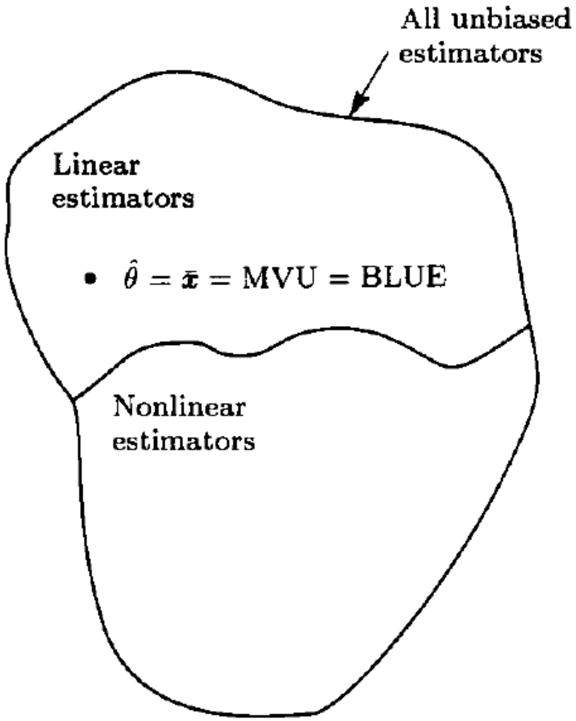
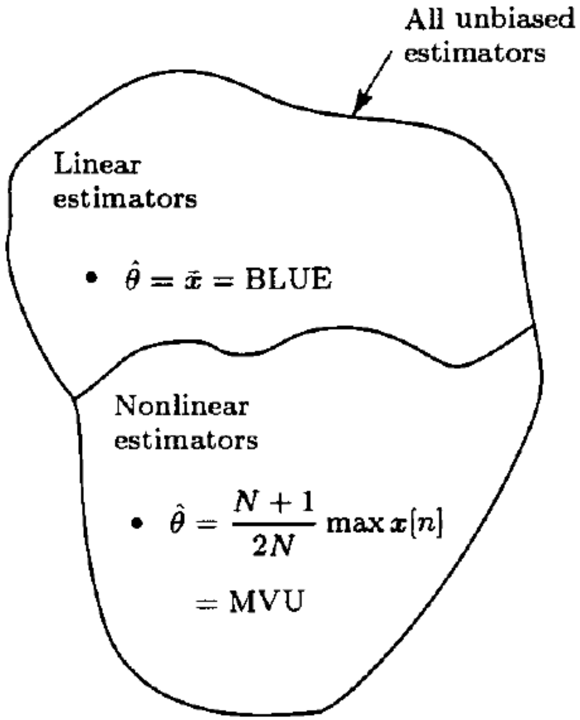
- The BLUE is an unbiased *linear* estimator with the minimum variance.

Linear estimator (linear in the data): $\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n]$.



(a) DC level in WGN; BLUE is optimal

$$\hat{\theta} = \bar{x} = \sum_{n=0}^{N-1} \frac{1}{N} x[n]$$



(b) Mean of uniform noise; BLUE is suboptimal

BLUE: a scalar case

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] \qquad \mathbf{a} = [a_0, a_1, \ldots, a_{N-1}]^T$$

Unbiased: $$E(\hat{\theta}) = \sum_{n=0}^{N-1} a_n E(x[n]) = \theta$$

$$E(x[n]) = s[n]\theta$$: linear function

$$x[n] = \theta s[n] + w[n]$$ noise

$$w[n] = x[n] - \mathbb{E}(x[n])$$

$$\begin{aligned}
\sum_{n=0}^{N-1} a_n E(x[n]) &= \theta \\
\sum_{n=0}^{N-1} a_n s[n]\theta &= \theta \\
\sum_{n=0}^{N-1} a_n s[n] &= 1
\end{aligned}$$

$$\mathbf{a}^T\mathbf{s} = 1$$

$$\begin{aligned}
\operatorname{var}(\hat{\theta}) &= E\left[\left(\mathbf{a}^T\mathbf{x} - \mathbf{a}^T E(\mathbf{x})\right)^2\right] \\
&= E\left[\left(\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))\right)^2\right] \\
&= E\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\mathbf{a}\right] \\
&= \mathbf{a}^T\mathbf{C}\mathbf{a}.
\end{aligned}$$

BLUE: $$\arg\min\left(\mathbf{var}(\hat{\theta}) = \mathbf{a}^T C\mathbf{a}\right) \quad \text{subject to } \mathbf{a}^T\mathbf{s} = 1.$$

BLUE: a scalar case

BLUE: $\arg\min\left(\mathbf{var}(\hat{\theta}) = \mathbf{a}^T C \mathbf{a}\right)$ subject to $\mathbf{a}^T \mathbf{s} = 1.$

$$J = \mathbf{a}^T \mathbf{C} \mathbf{a} + \lambda(\mathbf{a}^T \mathbf{s} - 1)$$

$\dfrac{\partial J}{\partial \mathbf{a}} = 2\mathbf{C}\mathbf{a} + \lambda \mathbf{s}$ (set to 0) => $\mathbf{a} = -\dfrac{\lambda}{2}\mathbf{C}^{-1}\mathbf{s}$ $\quad \mathbf{a}^T \mathbf{s} = -\dfrac{\lambda}{2}\mathbf{s}^T \mathbf{C}^{-1}\mathbf{s} = 1$

$$\mathbf{a}_{\text{opt}} = \frac{\mathbf{C}^{-1}\mathbf{s}}{\mathbf{s}^T \mathbf{C}^{-1}\mathbf{s}} \qquad -\frac{\lambda}{2} = \frac{1}{\mathbf{s}^T \mathbf{C}^{-1}\mathbf{s}}$$

$$\begin{aligned} \text{var}(\hat{\theta}) &= \mathbf{a}_{\text{opt}}^T \mathbf{C} \mathbf{a}_{\text{opt}} \\ &= \frac{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{C} \mathbf{C}^{-1} \mathbf{s}}{(\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s})^2} \\ &= \frac{1}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}}. \end{aligned}$$

BLUE: $\hat{\theta} = \mathbf{a}_{opt}^T \mathbf{x} = \dfrac{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{x}}{\mathbf{s}^T \mathbf{C}^{-1} \mathbf{s}}$

BLUE: a scalar case example

$$x[n] = A + w[n], \quad \text{where } w[n] \sim \mathcal{N}(0, \sigma_n^2)$$

$$E(x[n]) = s[n]\theta$$ => s = 1

BLUE:
$$\begin{aligned} \hat{A} &= \frac{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{x}}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}} \\ \operatorname{var}(\hat{A}) &= \frac{1}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}}. \end{aligned}$$

$$\mathbf{C} = \begin{bmatrix} \sigma_0^2 & 0 & \dots & 0 \\ 0 & \sigma_1^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{N-1}^2 \end{bmatrix}$$

$$\hat{A} = \frac{\sum_{n=0}^{N-1} \frac{x[n]}{\sigma_n^2}}{\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2}}$$

$$\operatorname{var}(\hat{A}) = \frac{1}{\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2}}.$$

If $\sigma^2 = \sigma_n^2$ for all n,

$$\begin{aligned} \hat{A} &= \frac{\mathbf{1}^T \frac{1}{\sigma^2} \mathbf{I} \mathbf{x}}{\mathbf{1}^T \frac{1}{\sigma^2} \mathbf{I} \mathbf{1}} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} x[n] = \bar{x} \\ \operatorname{var}(\hat{A}) &= \frac{1}{\mathbf{1}^T \frac{1}{\sigma^2} \mathbf{1}} \\ &= \frac{\sigma^2}{N}. \end{aligned}$$

BLUE: a vector case

$$\hat{\boldsymbol{\theta}} = \mathbf{A}\mathbf{x} \qquad \hat{\theta} \in \mathbb{R}^p,\ A \in \mathbb{R}^{p\times N}.$$

Unbiased: $E(\hat{\boldsymbol{\theta}}) = \mathbf{A}E(\mathbf{x}) = \boldsymbol{\theta}$

$$E(\mathbf{x}) = \mathbf{H}\boldsymbol{\theta} \qquad \mathbf{H} \in \mathbb{R}^{N\times p}$$

$$E(\mathbf{x}) = \underbrace{\begin{bmatrix} s[0] \\ s[1] \\ \vdots \\ s[N-1] \end{bmatrix}}_{\mathbf{H}} \theta \qquad \mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_p^T \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \dots & \mathbf{h}_p \end{bmatrix}$$

$$\mathbf{AH} = \mathbf{I} \quad \Rightarrow \quad \mathbf{a}_i^T \mathbf{h}_j = \delta_{ij} \qquad i = 1, 2, \dots, p;\ j = 1, 2, \dots, p$$

Appendix 6B

BLUE: $\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x}$

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}$$

Theorem 6.1 (Gauss-Markov Theorem) *If the data are of the general linear model form*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{6.18}$$

where $\mathbf{H}$ *is a known* $N \times p$ *matrix,* $\boldsymbol{\theta}$ *is a* $p \times 1$ *vector of parameters to be estimated, and* $\mathbf{w}$ *is a* $N \times 1$ *noise vector with zero mean and covariance* $\mathbf{C}$ *(the PDF of* $\mathbf{w}$ *is otherwise arbitrary), then the BLUE of* $\boldsymbol{\theta}$ *is*

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x} \tag{6.19}$$

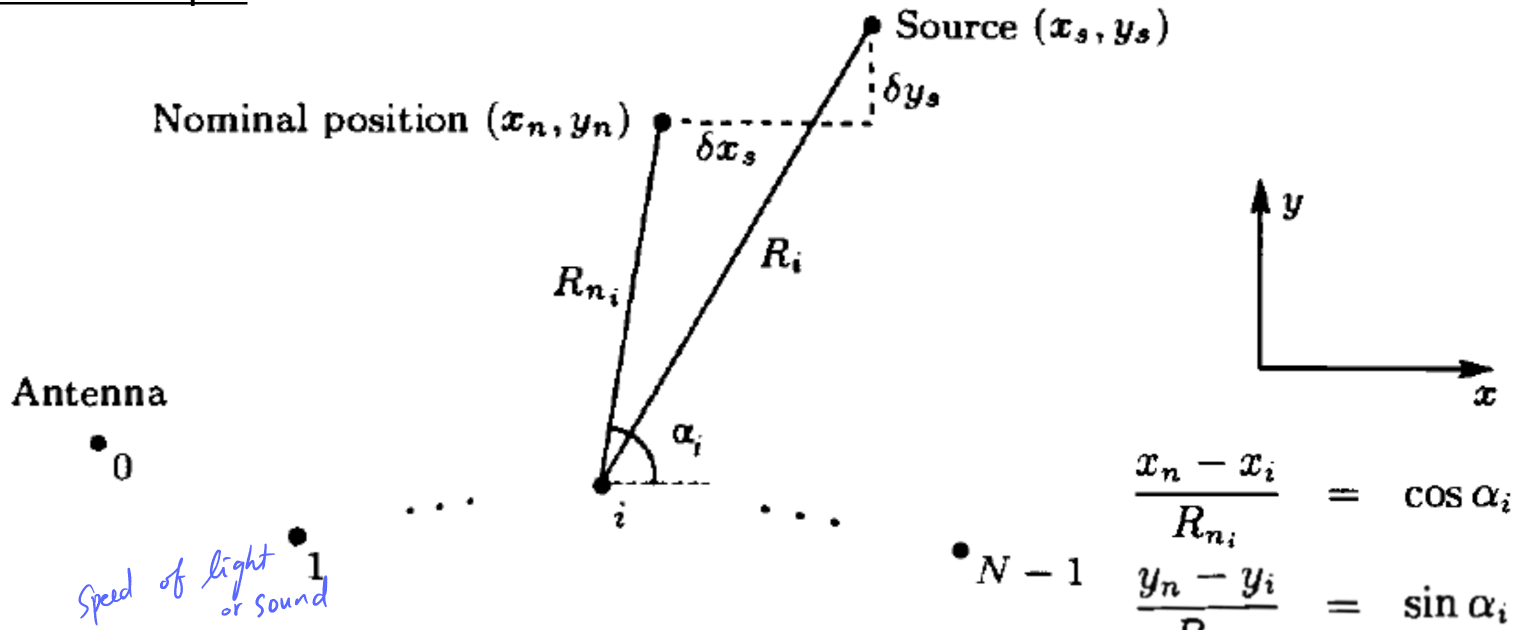
and the minimum variance of $\hat{\theta}_i$ *is*

$$\mathrm{var}(\hat{\theta}_i) = \left[(\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\right]_{ii}. \tag{6.20}$$

In addition, the covariance matrix of $\hat{\boldsymbol{\theta}}$ *is*

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}. \tag{6.21}$$

Source Localization Example



$$\frac{x_n - x_i}{R_{n_i}} = \cos\alpha_i \qquad \frac{y_n - y_i}{R_{n_i}} = \sin\alpha_i$$

Speed of light or sound

$$t_i = T_0 + R_i/c + \epsilon_i \qquad i = 0, 1, \ldots, N-1$$

$$R_i = \sqrt{(x_s - x_i)^2 + (y_s - y_i)^2} \qquad R_i \approx R_{n_i} + \frac{x_n - x_i}{R_{n_i}}\delta x_s + \frac{y_n - y_i}{R_{n_i}}\delta y_s$$

; 1st order Taylor expansion

$$t_i = T_0 + \frac{R_{n_i}}{c} + \frac{x_n - x_i}{R_{n_i} c}\delta x_s + \frac{y_n - y_i}{R_{n_i} c}\delta y_s + \epsilon_i$$

$$t_i = T_0 + \frac{R_{n_i}}{c} + \frac{\cos\alpha_i}{c}\delta x_s + \frac{\sin\alpha_i}{c}\delta y_s + \epsilon_i$$

$$\tau_i = T_0 + \frac{\cos\alpha_i}{c}\delta x_s + \frac{\sin\alpha_i}{c}\delta y_s + \epsilon_i \qquad \tau_i = t_i - \frac{R_{n_i}}{c}$$

Source Localization Example

$$\tau_i = T_0 + \frac{\cos\alpha_i}{c}\delta x_s + \frac{\sin\alpha_i}{c}\delta y_s + \epsilon_i \qquad \text{Unknowns: } (T_0, \delta x_s, \delta y_s)$$

Time difference of arrival (TDOA):

$$\begin{aligned} \xi_1 &= \tau_1 - \tau_0 \\ \xi_2 &= \tau_2 - \tau_1 \\ &\vdots \\ \xi_{N-1} &= \tau_{N-1} - \tau_{N-2} \end{aligned}$$

$$\xi_i = \frac{1}{c}(\cos\alpha_i - \cos\alpha_{i-1})\delta x_s + \frac{1}{c}(\sin\alpha_i - \sin\alpha_{i-1})\delta y_s + \epsilon_i - \epsilon_{i-1}$$

$$\mathbf{w} = \underbrace{\begin{bmatrix} -1 & 1 & 0 & 0 & \dots & 0 \\ 0 & -1 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & -1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \epsilon_0 \\ \epsilon_1 \\ \vdots \\ \epsilon_{N-1} \end{bmatrix}}_{\boldsymbol{\epsilon}}$$

$$\boldsymbol{\theta} = \begin{bmatrix} \delta x_s & \delta y_s \end{bmatrix}^T$$

$$\mathbf{H} = \frac{1}{c}\begin{bmatrix} \cos\alpha_1 - \cos\alpha_0 & \sin\alpha_1 - \sin\alpha_0 \\ \cos\alpha_2 - \cos\alpha_1 & \sin\alpha_2 - \sin\alpha_1 \\ \vdots & \vdots \\ \cos\alpha_{N-1} - \cos\alpha_{N-2} & \sin\alpha_{N-1} - \sin\alpha_{N-2} \end{bmatrix}$$

$$\mathbf{w} = \begin{bmatrix} \epsilon_1 - \epsilon_0 \\ \epsilon_2 - \epsilon_1 \\ \vdots \\ \epsilon_{N-1} - \epsilon_{N-2} \end{bmatrix}.$$

$$\mathbf{C} = E[\mathbf{A}\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T\mathbf{A}^T] = \sigma^2\mathbf{A}\mathbf{A}^T$$

Variance

Cov만 알아내면
BLUE에 적용하면 됨.

BLUE:
$$\begin{aligned} \hat{\boldsymbol{\theta}} &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\boldsymbol{\xi} \\ &= [\mathbf{H}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{H}]^{-1}\mathbf{H}^T(\mathbf{A}\mathbf{A}^T)^{-1}\boldsymbol{\xi} \end{aligned}$$

$$\mathbf{C}_{\hat{\theta}} = \sigma^2\left[\mathbf{H}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{H}\right]^{-1}$$